# Persistence Diagrams as Barycenters for Wassertein Distances

**Report on *Wasserstein Dictionaries of Persistence Diagrams***


Matthieu Pierre Boyer*
École Normale Supérieure
Paris, France
matthieu.boyer@ens.fr

Gary Klajer*
ÉNS Paris-Saclay
Gif-sur-Yvette, France
gary.klajer@ens-paris-saclay.fr

Marc Perlade*
École Normale Supérieure
Paris, France
marc.perlade@ens.fr


**Figure 1**


## Abstract

In this report based on [SDT23], we study the idea of doing non-linear dimensionality reduction on a large dataset of persistence diagrams, using atomic persistence diagrams that span the dataset for the hull defined by the optimal transport of diagrams. We will first give the mathematical intuition on why this could work then provide an algorithm that proposes a method to compute such a representation. We will finally provide benchmarks on the different methods, implemented in Rust.



## Keywords

Persistence, Dictionaries, Optimization, Topological Data Analysis


## Introduction

## 1 Mathematical preliminaries

### 1.1 Representation of persistence diagrams

We are given a set $X = \{X_1, \ldots, X_n\}$ of $N$ persistence diagrams. Each input diagram is given as the set of its persistence pairs mapped to points in the 2-dimensional plane[1], along with their birth and death persistence.

### 1.2 2-dimensional optimal transport

Optimal transport is the mathematical field concerned with mapping similarities from a space with some sense of geometry to the space of probability distributions on that space.

### 1.3

## 2 Algorithm

### 2.1

### 2.2 Implementation

## 3 Analysis and comparison of implementations

## References


[SDT23] Keanu Sisouk, Julie Delon, and Julien Tierny. Wasserstein dictionaries of persistence diagrams, 2023.


*All authors contributed equally.

[1] Or half-plane above the line $y = x$.